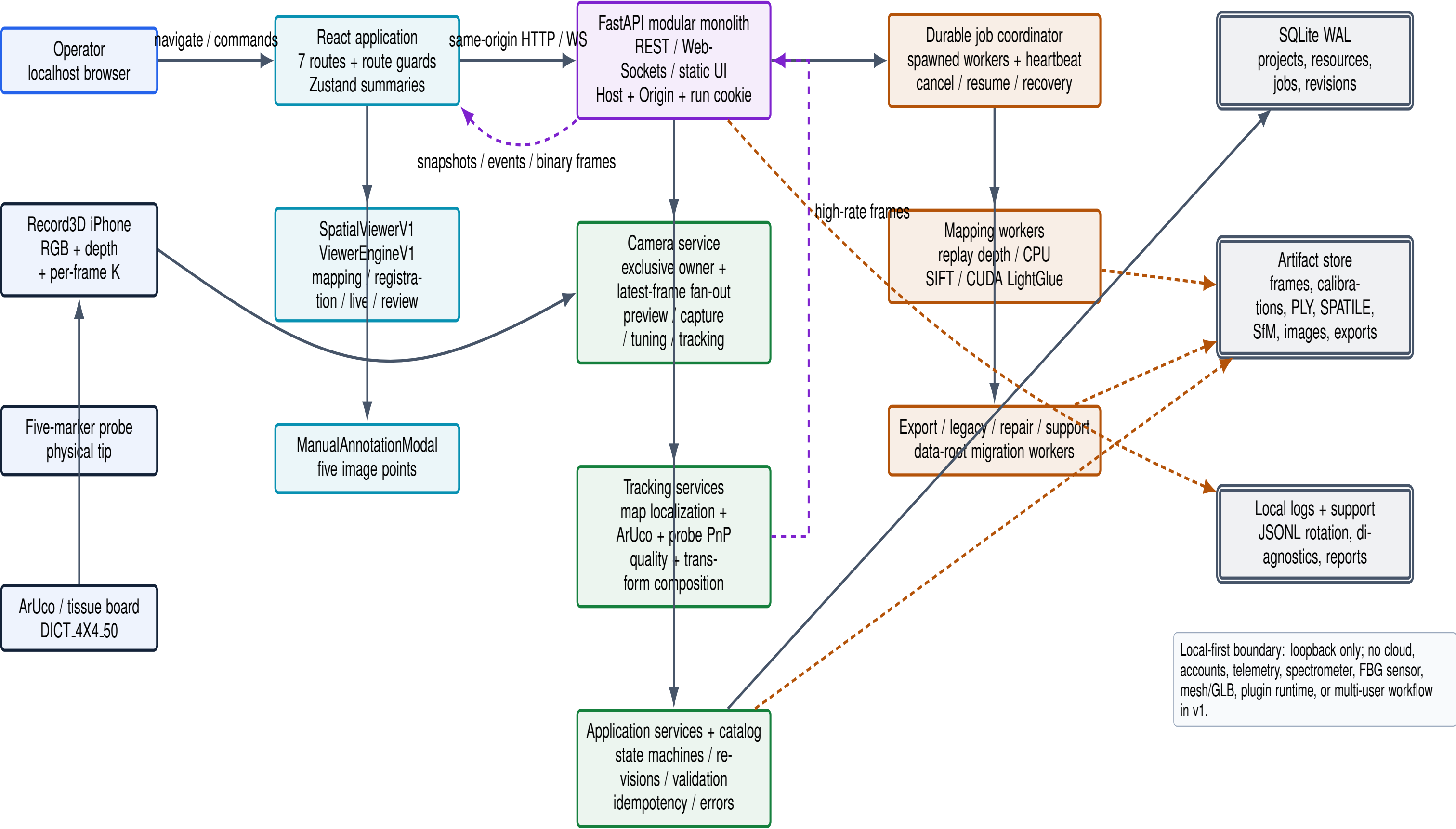
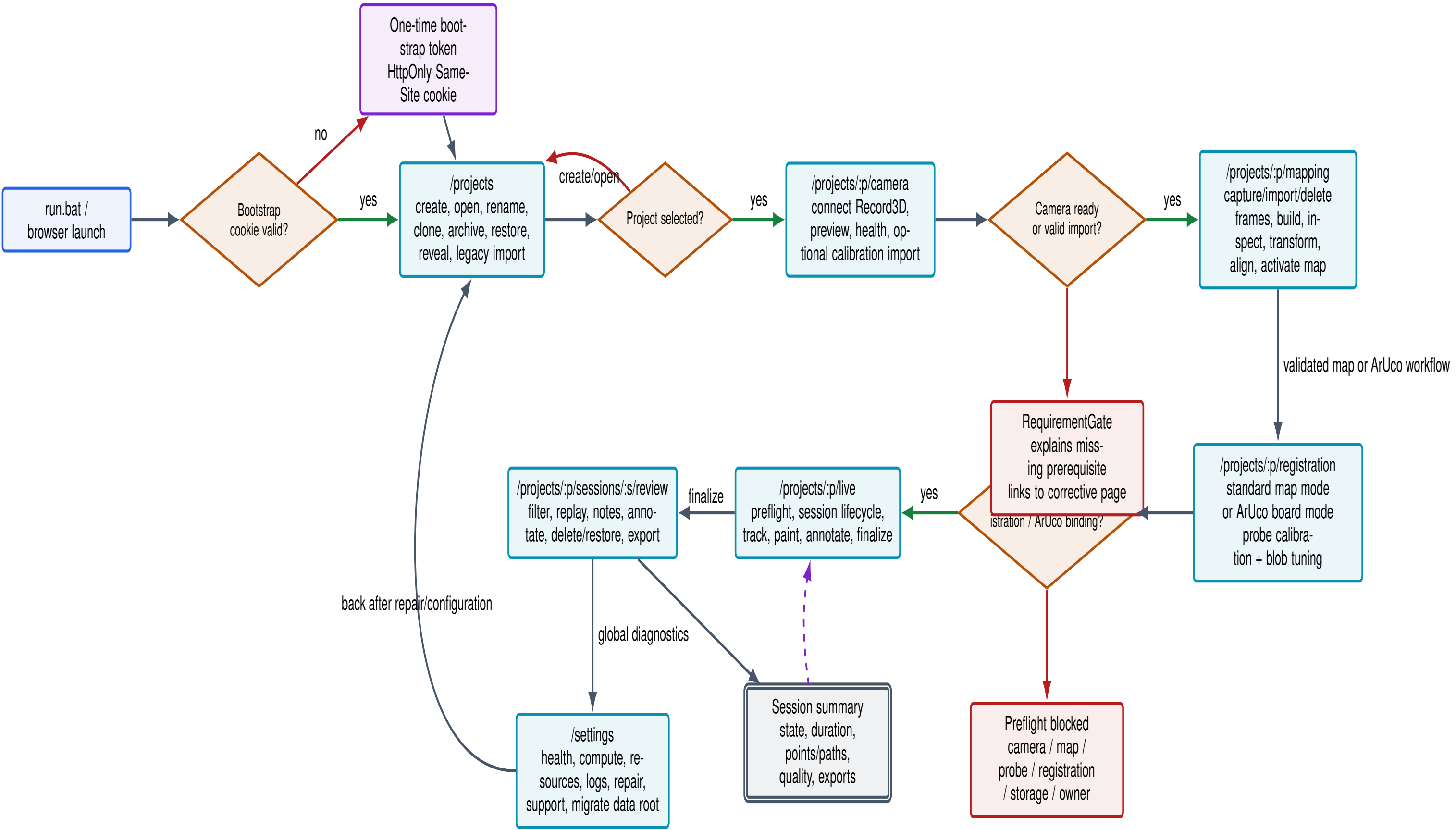


1. End-to-end system and responsibility flow

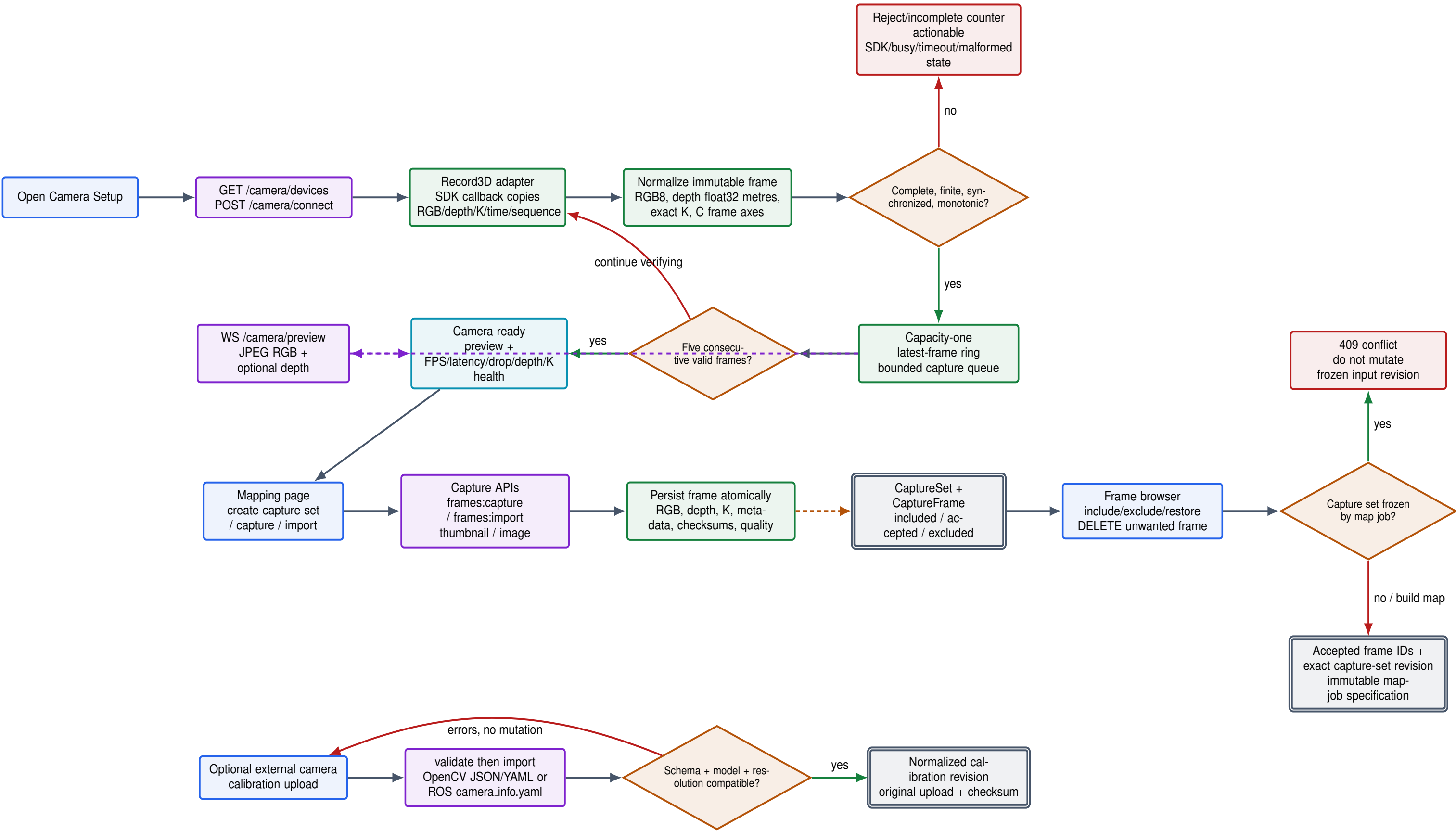
Legend: User/UI API/WS Service/Pipeline Worker Durable data Gap/failure





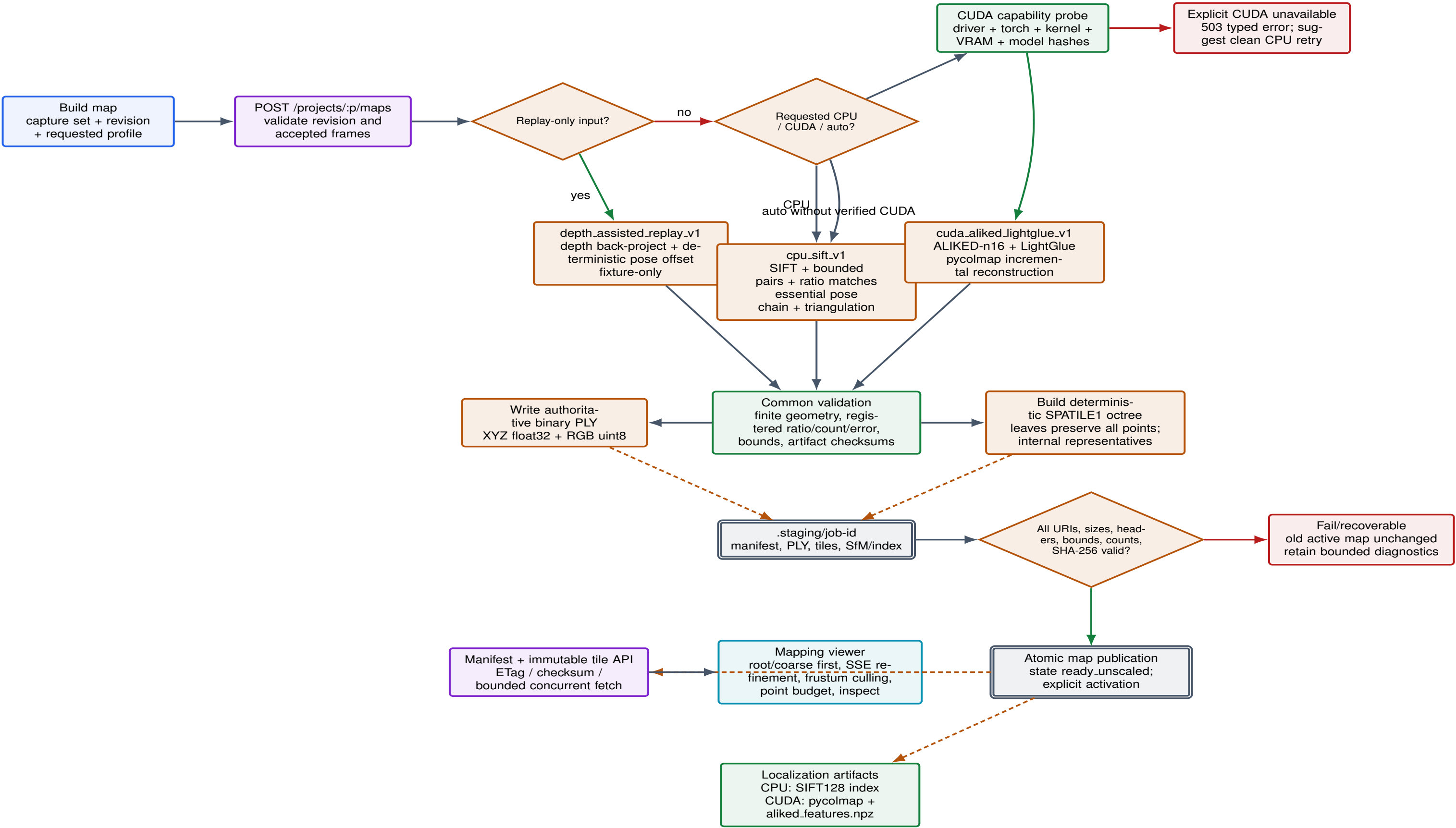
3. Camera setup, normalized frames, capture sets, and frame management

Record3D is primary; external calibration is import-only



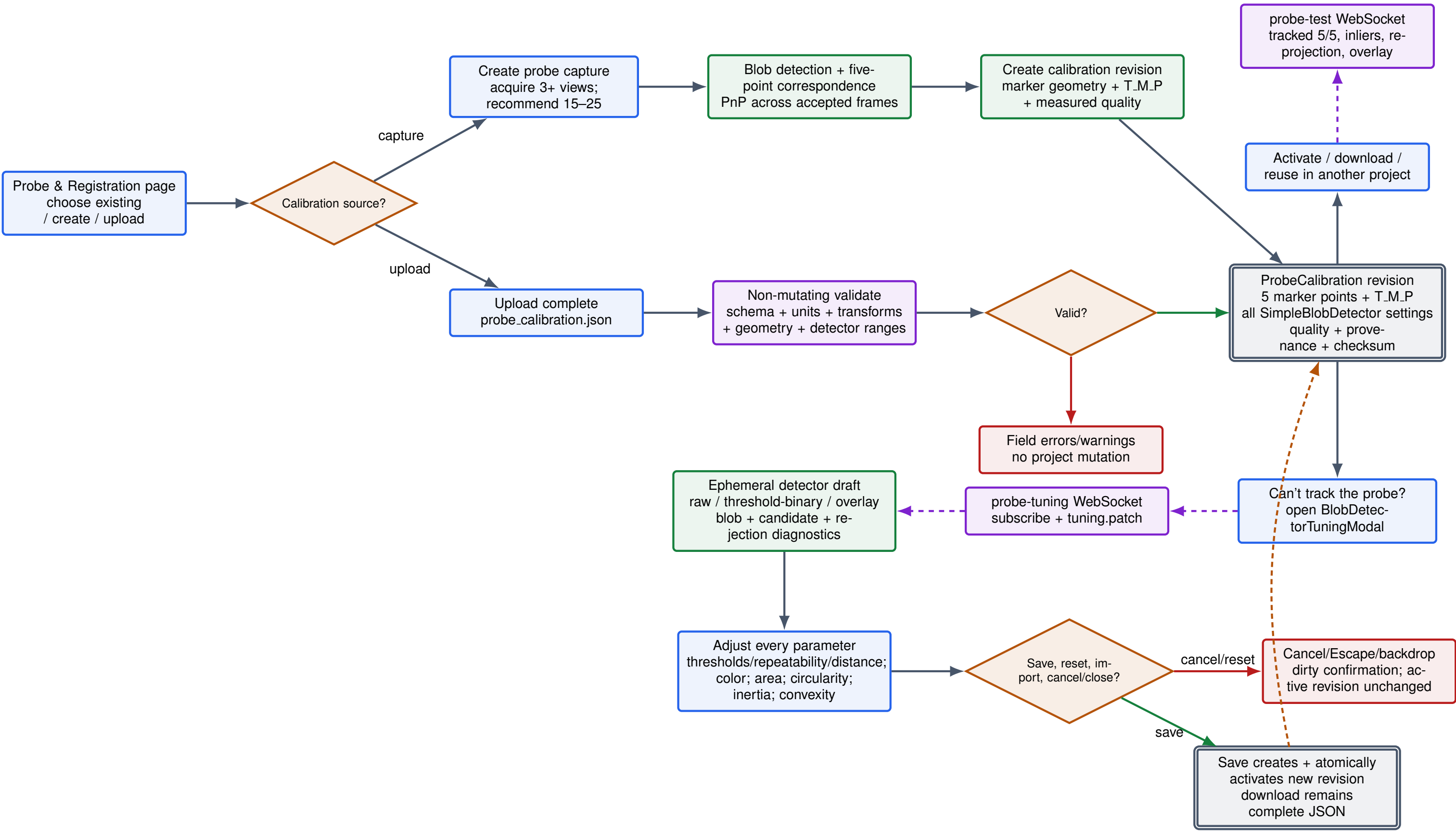
4. Mapping dispatch, reconstruction profiles, publication, and browser LOD

No silent algorithm substitution inside an attempt



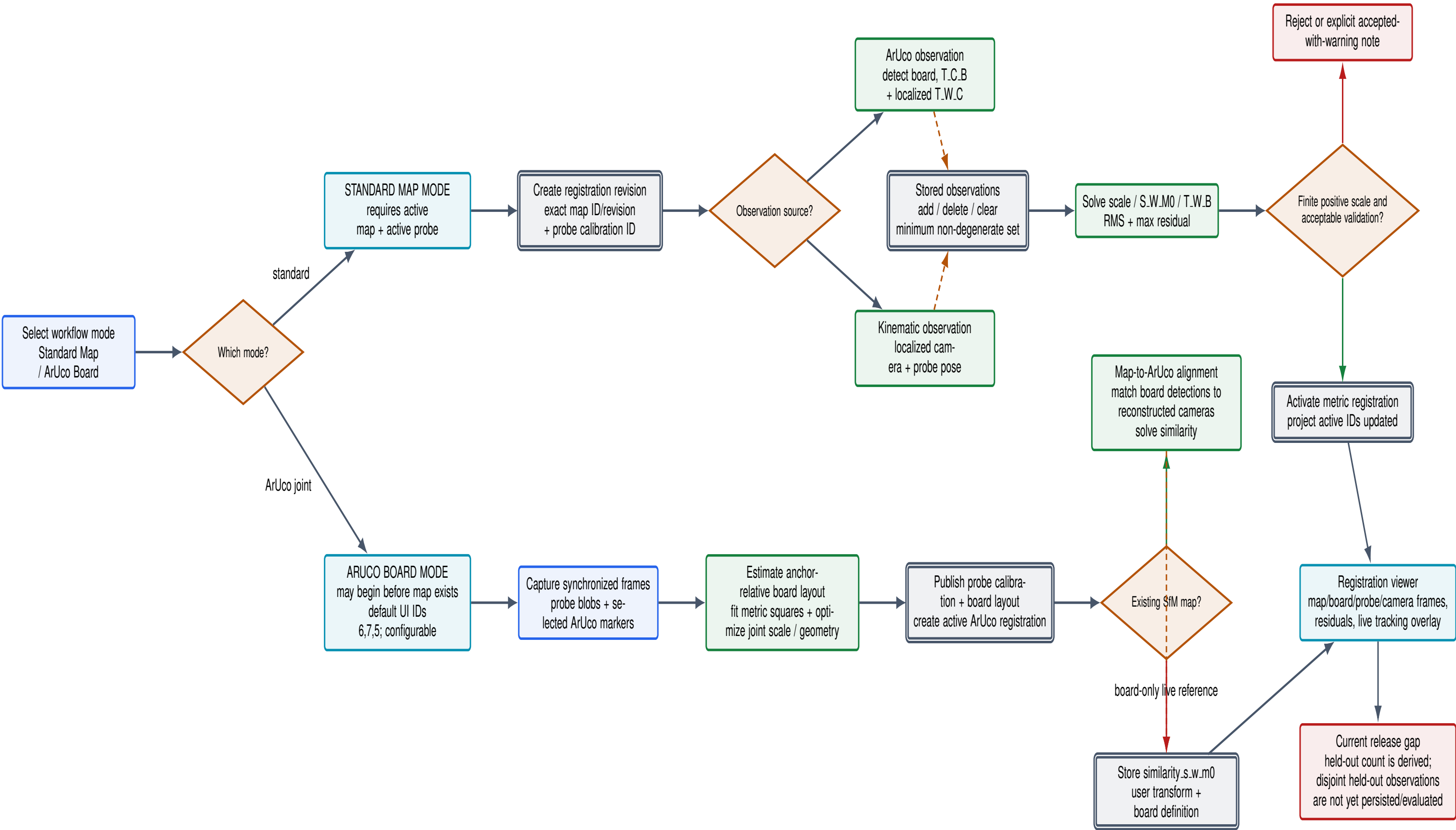
5. Probe calibration, reusable JSON, tracking test, and blob-detector tuning

The “Can’t track the probe?” subsection is modal and draft-safe



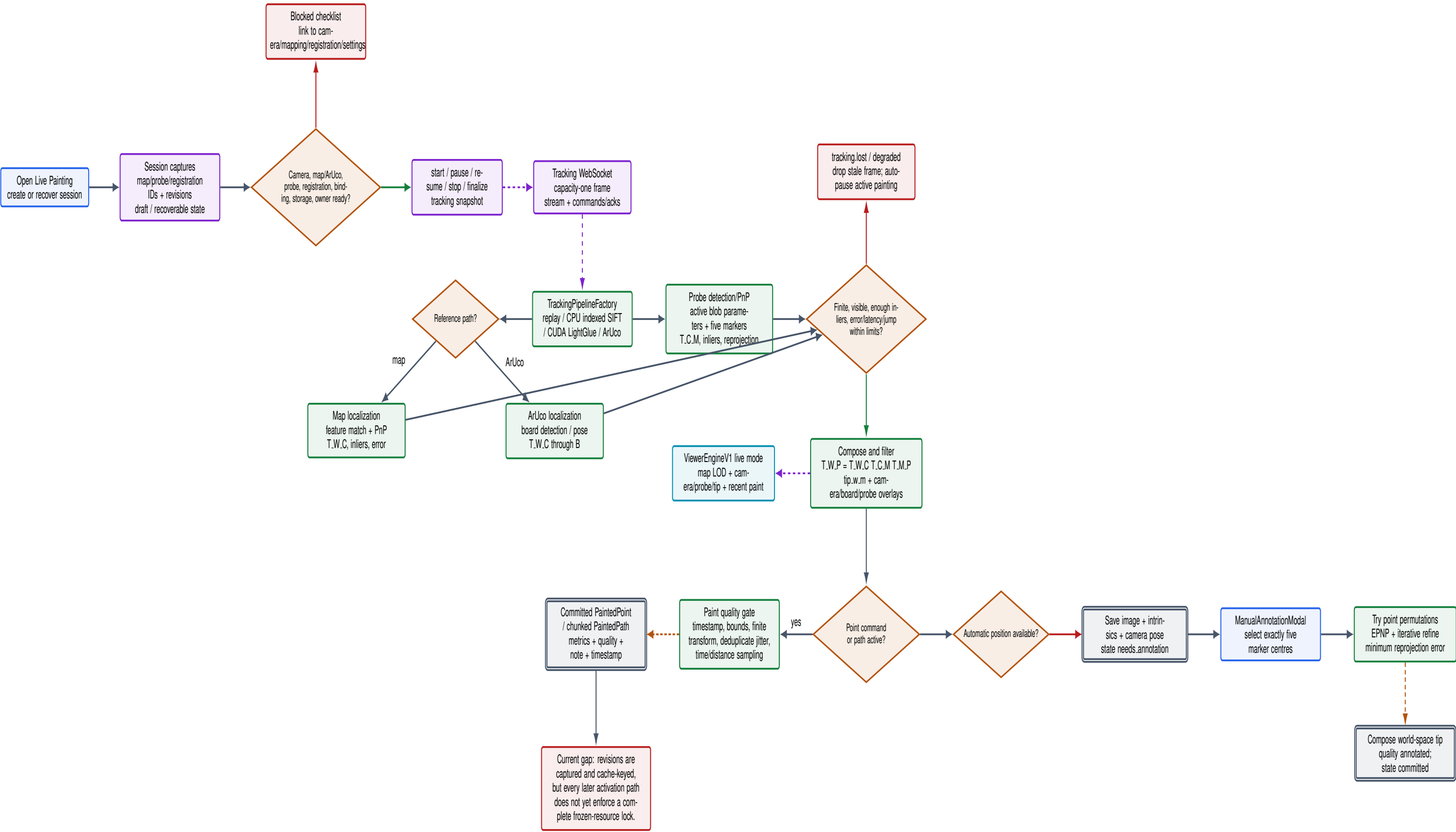
6. Registration logic: standard map mode and ArUco board mode

Both routes bind probe geometry to a physical reference frame



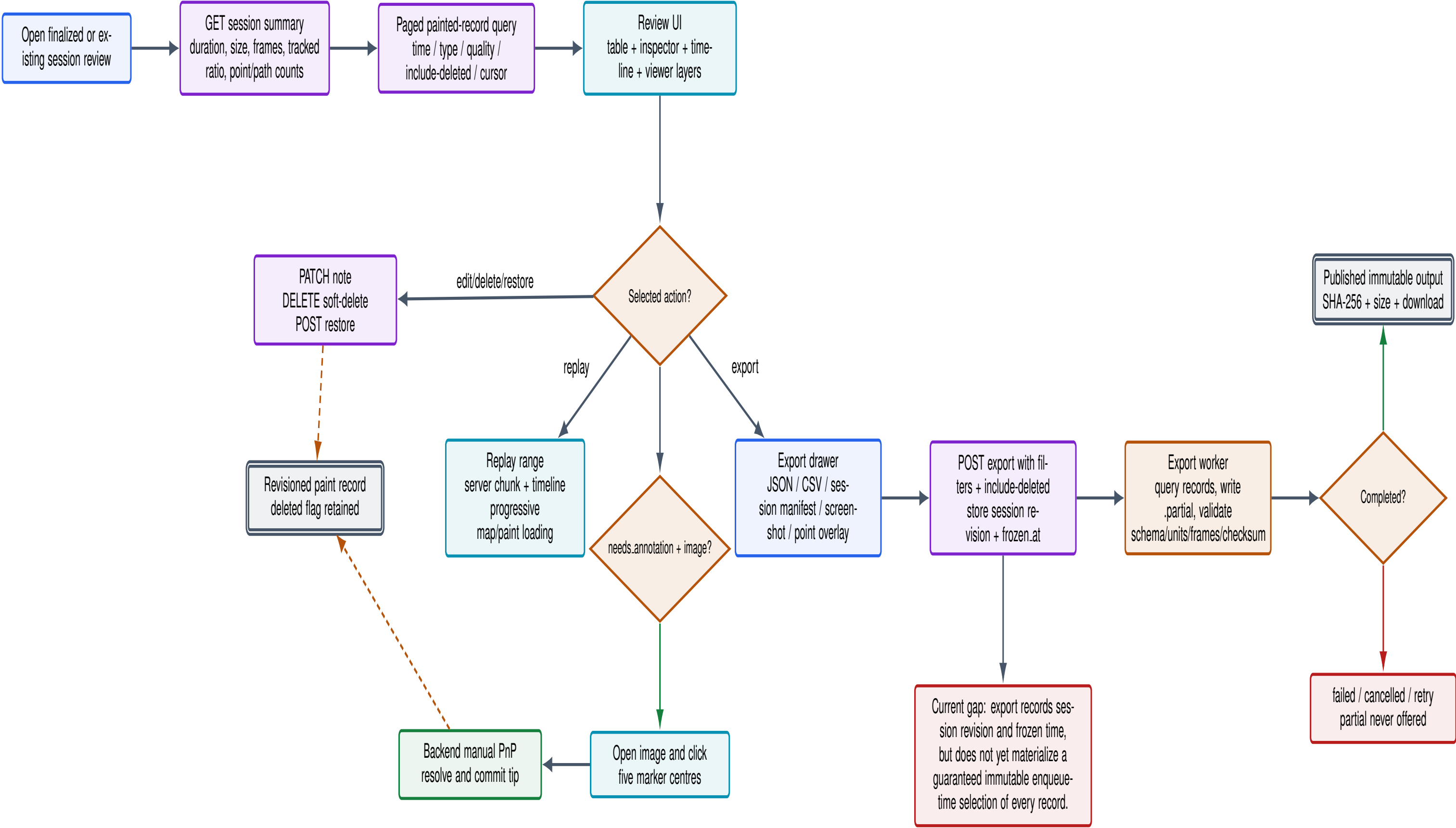
7. Live session preflight, localization, probe-tip composition, and painting

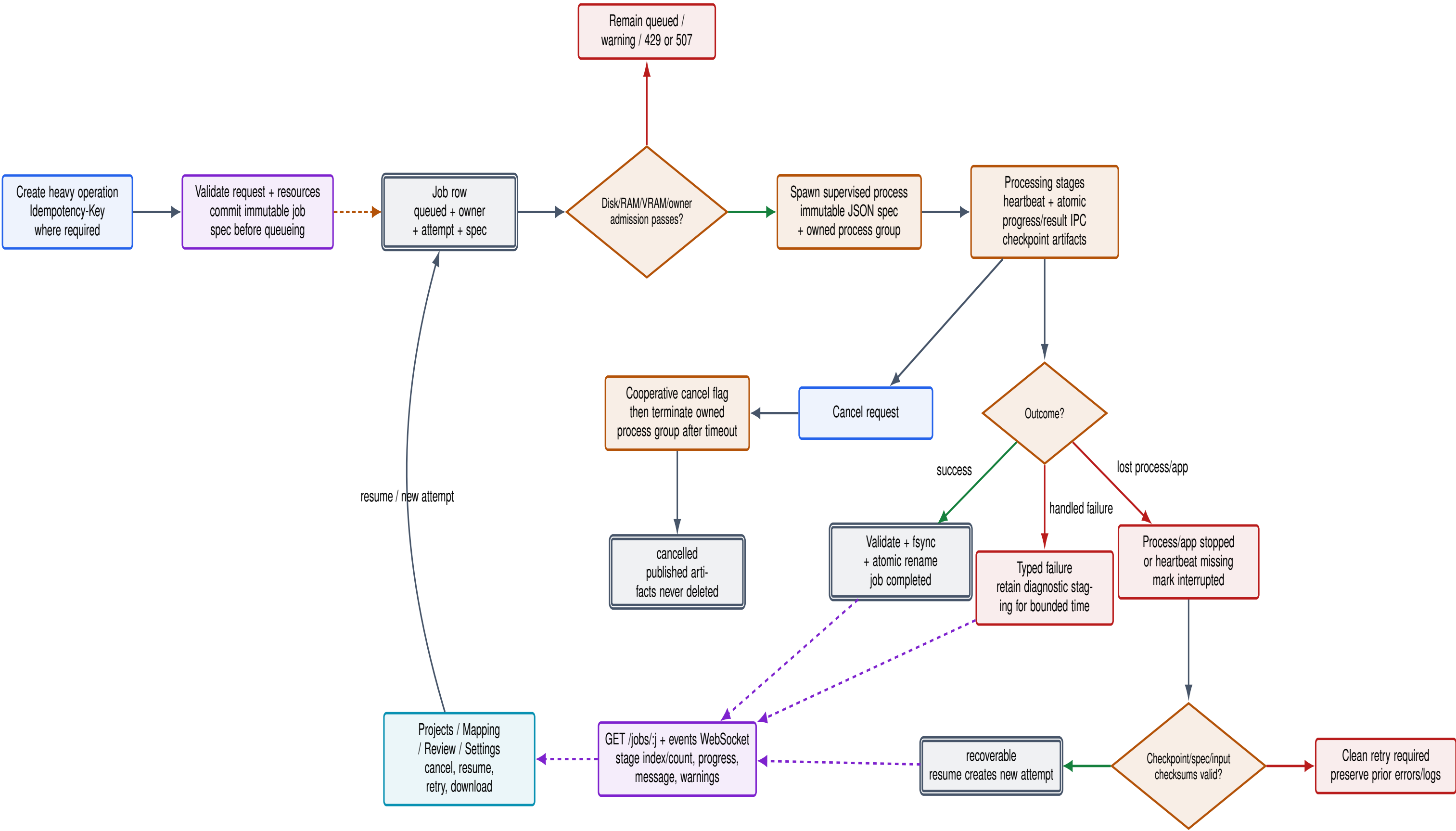
High-rate frames are ephemeral; paint records are durable

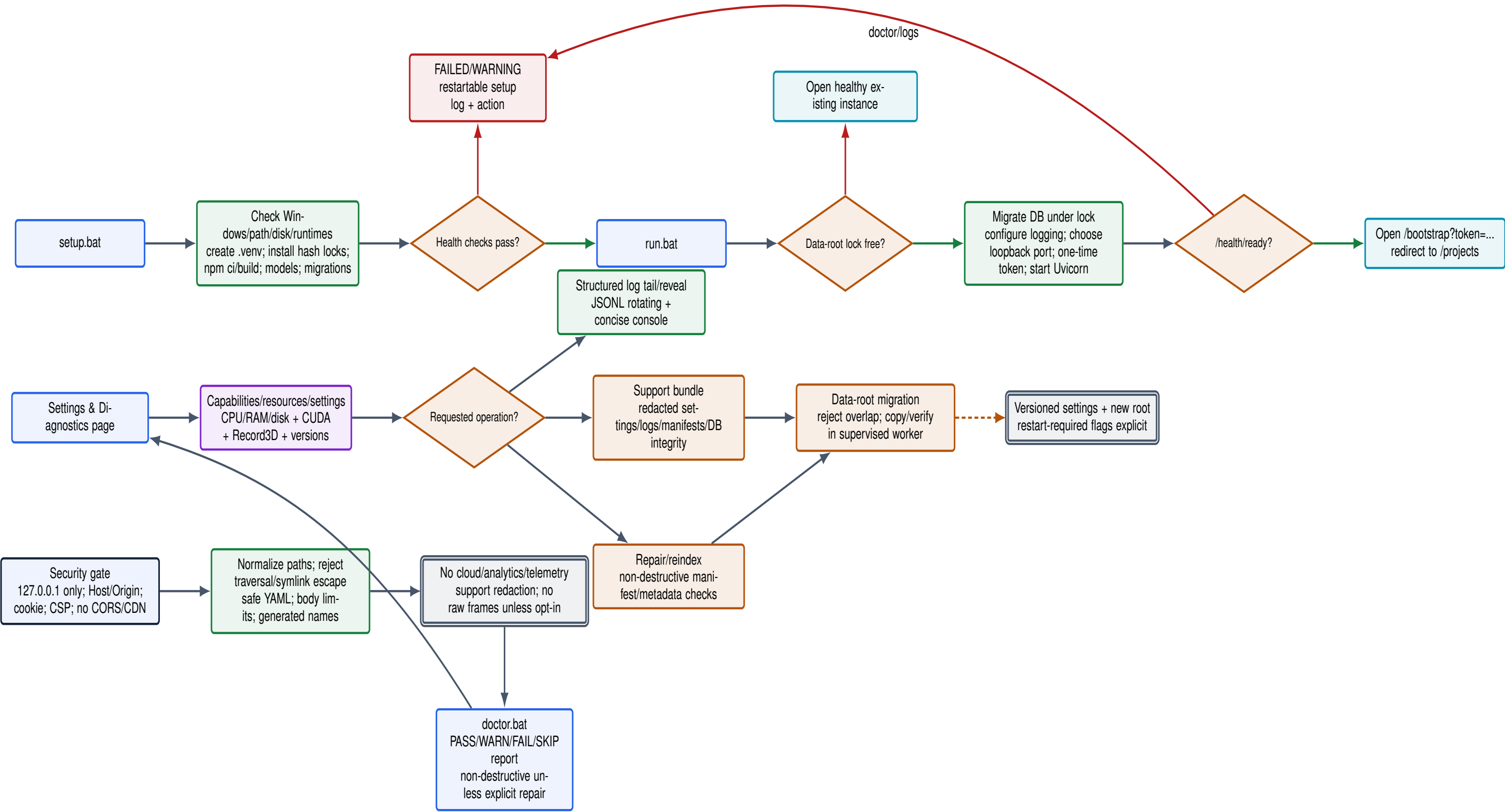


8. Session review, replay, record editing, manual annotation, and exports

Server records remain authoritative

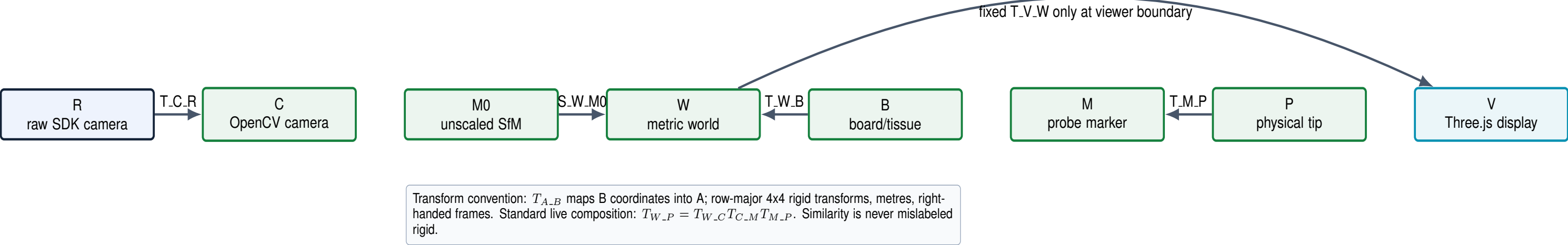
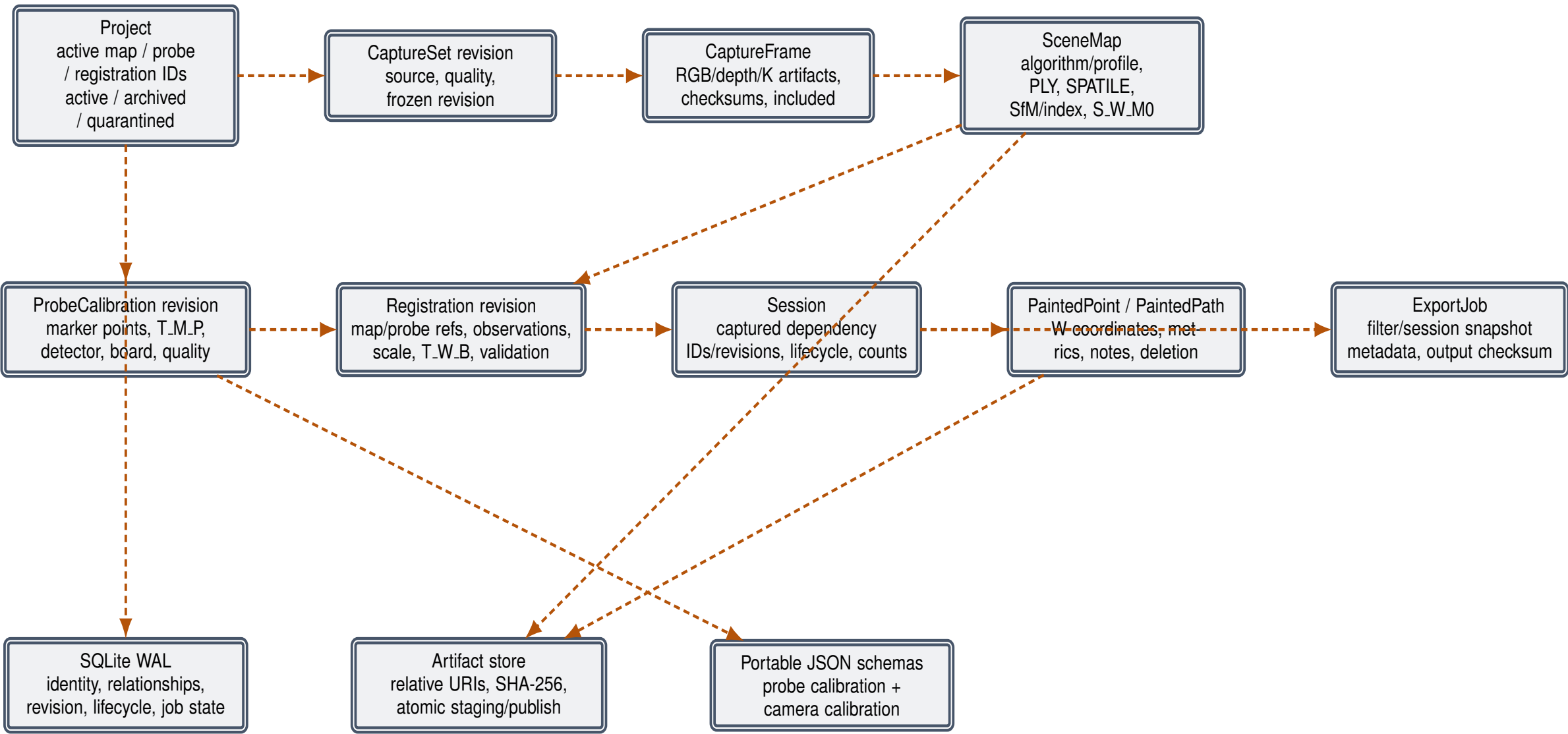






11. Persistence, domain relationships, revisions, and coordinate transforms

SQLite owns identity/state; files own large immutable artifacts



12. API, WebSocket, and function coverage index

Every current user-visible domain is represented in pages 1–11

Domain	REST functions	Stream / internal logic	Flow pages
System	live/ready health; capabilities; resources; diagnostics; settings get/patch	startup lock, CUDA probe, logging, resource admission	1, 9, 10
Projects	list/create/get/rename; summary; clone; archive/restore; reveal; jobs	project selection, readiness, warnings, legacy import	2, 9, 11
Camera	devices/status; connect/disconnect	preview WS; exclusive owner; normalized capacity-one frames	3
External calibration	validate/import/get/activate	schema/model/resolution compatibility; preserve original	3
Capture	sets list/create/get; capture/import; include patch; delete; thumbnail/image	quality, checksums, revision freeze	3, 4
Maps	list/create/get; transform; align-aruco; activate; manifest/tiles	replay/CPU/CUDA mapping; SPATILE LOD; localization artifacts	4, 6, 11
Probe calibration	capture/create; validate/import/get/download/activate/revise	probe-test and probe-tuning WS; blob detection; PnP; complete JSON	5
ArUco calibration	capture/solve; map align	joint board/probe observations; layout and scale optimization	6
Registration	list/create/get; add/delete/clear observations; solve/validate/activate	registration tracking WS; map/kinematic/ArUco transforms	6
Sessions	list/create/get; start/pause/resume/stop/finalize; snapshot	tracking WS; frozen refs; lifecycle/reconnect/recovery	7, 11
Painting	point/path create; recent tracking; undo	quality gate; time/distance sampling; chunks; image fallback	7
Review	paged records/points/paths; replay; note; delete/restore; image; annotate	viewer filters/timeline; five-point manual PnP	8
Exports	list/create/download	JSON, CSV, session manifest, screenshot, point overlay; checksum publication	8, 9
Jobs	get/cancel/resume	events WS; spawn, heartbeat, progress, checkpoint, retry/recovery	9
Operations	support bundle; repair/reindex; data-root migration; log tail/reveal	supervised operations worker; redaction; overlap checks	9, 10
Legacy migration	create/status/report	staged import, normalization, validation, untouched source	1, 9
Viewer	map manifest/tiles and frame images	mapping, registration, live, review modes; LOD; transforms; overlays; disposal	1, 2, 4, 6–8

Release gap: replay tuple depth is incompatible with the byte-buffer-only capture persistence path; two replay workflow tests fail.

Release gap: CPU mapping is the shipped essential-matrix pose-chain implementation, not global pycolmap bundle adjustment.

Release gap: disjoint held-out registration evidence and complete frozen-session dependency enforcement remain open.

Release gap: some diagnostics/replay metrics are simulated and final physical probe geometry requires hardware validation.

Release gap: exports do not yet guarantee a materialized immutable enqueue-time record selection.

Release gate: Record3D, CUDA, installer, full browser E2E, and target-machine performance need tagged validation.